Chocolate Sales Report

1. Project Overview

This project focuses on analysing chocolate sales data to understand sales performance across different countries, products, and salespersons. The dataset used in this project was obtained from Kaggle.

Python was used to inspect the dataset, verify data types, and perform basic sales analysis. After preparing the data, a sales dashboard was developed using Power BI to visualize key sales metrics. The dashboard helps present information such as total sales, product performance, and regional sales distribution in a clear and interactive format.

2. Dataset Summary

The chocolate sales dataset contains information about product sales transactions. Each record in the dataset represents a single sales entry.

The dataset contains 3,282 rows and 6 columns. The columns included in the dataset are:

- Sales Person – Name of the salesperson responsible for the sale
- Country – Country where the product was sold
- Product – Type of chocolate product sold
- Date – Date when the sale occurred
- Amount – Total sales value of the transaction
- Boxes Shipped – Number of boxes shipped in the order

The dataset was examined in Python to understand the data structure and confirm the number of rows, columns, and data types.

3. Exploratory Data Analysis using python

Exploratory analysis was performed using Python and the Pandas library. The dataset was loaded into a Data Frame and basic inspection commands such as head() and info() were used to view the data and understand its structure.

```
[5]: import pandas as pd
     df=pd.read_csv('Chocolate Sales-2.csv')

[6]: df.head()
```

[6]:

	Sales Person	Country	Product	Date	Amount	Boxes Shipped
0	Jehu Rudeforth	UK	Mint Chip Choco	04-01-2022	$5,320.00	180
1	Van Tuxwell	India	85% Dark Bars	01-08-2022	$7,896.00	94
2	Gigi Bohling	India	Peanut Butter Cubes	07-07-2022	$4,501.00	91
3	Jan Morforth	Australia	Peanut Butter Cubes	27-04-2022	$12,726.00	342
4	Jehu Rudeforth	UK	Peanut Butter Cubes	24-02-2022	$13,685.00	184

```
[7]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3282 entries, 0 to 3281
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Sales Person   3282 non-null   object
 1   Country        3282 non-null   object
 2   Product        3282 non-null   object
 3   Date           3282 non-null   object
 4   Amount         3282 non-null   object
 5   Boxes Shipped  3282 non-null   int64
dtypes: int64(1), object(5)
memory usage: 154.0+ KB
```

The dataset was checked for missing values and the results showed that there were no missing entries. Column names were converted to lowercase for easier handling in Python.

```
[10]: # checking null values
      df.isnull().sum()

[10]: Sales Person     0
      Country          0
      Product          0
      Date             0
      Amount           0
      Boxes Shipped    0
      dtype: int64
```

During inspection, it was observed that the amount and date columns needed to be converted into appropriate data types. The amount column was converted into numeric format and the date column was converted into datetime format to support proper analysis.

```
[11]: df.columns=df.columns.str.lower()
      df.columns=df.columns.str.replace(' ','_')

[12]: df.columns

[12]: Index(['sales_person', 'country', 'product', 'date', 'amount',
             'boxes_shipped'],
            dtype='object')

[14]: # Convert Amount column to numeric
      df["amount"] = df["amount"].replace('[\$,]', '', regex=True).astype(float)

[20]: # Convert Date column
      df["date"] = pd.to_datetime(df["date"], dayfirst=True)

[21]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3282 entries, 0 to 3281
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   sales_person   3282 non-null   object
 1   country        3282 non-null   object
 2   product        3282 non-null   object
 3   date           3282 non-null   datetime64[ns]
 4   amount         3282 non-null   float64
 5   boxes_shipped  3282 non-null   int64
dtypes: datetime64[ns](1), float64(1), int64(1), object(3)
memory usage: 154.0+ KB
```

4. Data Analysis using python

After preparing the dataset, Python was used to perform sales analysis using grouping and aggregation functions.

The following analyses were performed:

• Total sales revenue was calculated from the amount column.
• Sales were grouped by country to analyse regional sales performance.
• Product-based analysis was conducted to identify the products generating the highest sales.
• Salesperson performance was analysed by grouping sales data by sales_person.
• The number of boxes shipped was analysed to understand shipment distribution across countries.

These analyses helped summarize the dataset and identify patterns in product sales, regional demand, and salesperson performance.

```
[24]: #performing sales anaysis
      #1. Total sales:
      df["amount"].sum()

[24]: np.float64(19791571.860000003)
```

```
[25]: #2. sales by country:
      df.groupby("country")["amount"].sum().sort_values(ascending=False)

[25]: country
      Australia      3646444.35
      UK             3365388.90
      India          3343730.83
      USA            3313858.09
      Canada         3078495.65
      New Zealand    3043654.04
      Name: amount, dtype: float64
```

```
[26]: #3. Top products:
      df.groupby("product")["amount"].sum().sort_values(ascending=False)

[26]: product
      Smooth Sliky Salty      1120201.09
      50% Dark Bites          1087659.35
      White Choc              1054257.00
      Peanut Butter Cubes     1036591.09
      Eclairs                  996947.96
      99% Dark & Pure          960033.45
      85% Dark Bars            955268.24
      Organic Choco Syrup      945345.73
      Spicy Special Slims      938131.69
      Mint Chip Choco          904989.55
      Almond Choco             890454.65
      Manuka Honey Choco       885208.03
      Milk Bars                866273.62
      Raspberry Choco          845388.41
      After Nines              837215.67
      Fruit & Nut Bars         829278.96
      Orange Choco             824070.85
      Drinking Coco            822245.57
      Baker's Choco Chips      799085.67
      Choco Coated Almonds     773062.91
      Caramel Stuffed Bars     742516.15
      70% Dark Bites           677346.22
      Name: amount, dtype: float64
```

[27]:
```
#4. Top sales person:
df.groupby("sales_person")["amount"].sum().sort_values(ascending=False)
```

[27]:
```
sales_person
Ches Bonnell           1022599.96
Oby Sorrel             1017204.12
Madelene Upcott        1010028.72
Kelci Walkden          1002929.10
Brien Boise             997326.48
Van Tuxwell             974425.09
Dennison Crosswaite     931849.57
Beverie Moffet          892421.37
Kaine Padly             849062.76
Marney O'Breen          836427.63
Barr Faughny            830826.91
Roddy Speechley         808359.58
Gunar Cockshoot         762283.43
Gigi Bohling            742296.51
Karlen McCaffrey        713832.64
Jehu Rudeforth          708505.03
Jan Morforth            706585.50
Curtice Advani          691797.92
Rafaelita Blaksland     668865.45
Husein Augar            656149.73
Andria Kimpton          648214.81
Mallorie Waber          642903.33
Camilla Castle          628613.13
Dotty Strutley          608101.17
Wilone O'Kielt          439961.92
Name: amount, dtype: float64
```

[32]:
```
#4.1. Top 10 sales person:
df.groupby("sales_person")["amount"].sum().sort_values(ascending=False).head(10)
```

[32]:
```
sales_person
Ches Bonnell           1022599.96
Oby Sorrel             1017204.12
Madelene Upcott        1010028.72
Kelci Walkden          1002929.10
Brien Boise             997326.48
Van Tuxwell             974425.09
Dennison Crosswaite     931849.57
Beverie Moffet          892421.37
Kaine Padly             849062.76
Marney O'Breen          836427.63
Name: amount, dtype: float64
```

•[28]:
```
#5. Boxes shipped by country:
df.groupby("country")["boxes_shipped"].sum().sort_values(ascending=False)
```

[28]:
```
country
Australia      99618
Canada         95158
UK             92523
India          89968
USA            81820
New Zealand    81350
Name: boxes_shipped, dtype: int64
```

•[33]:
```
#5.1. Top 3 Boxes shipped by country:
df.groupby("country")["boxes_shipped"].sum().sort_values(ascending=False).head(3)
```

[33]:
```
country
Australia    99618
Canada       95158
UK           92523
Name: boxes_shipped, dtype: int64
```

5. Chocolate Sales Dashboard

An interactive sales dashboard was created using Power BI to visualize the results of the analysis. The dashboard includes key visuals such as total sales, total boxes shipped, sales by country, sales by product, and sales by salesperson. These visualizations help present the sales data in a clear and understandable format.

A country slicer was added to allow users to filter the dashboard based on specific countries. This enables users to view and compare sales performance across different regions. The dashboard provides a quick overview of sales performance and helps highlight important trends in the chocolate sales dataset.